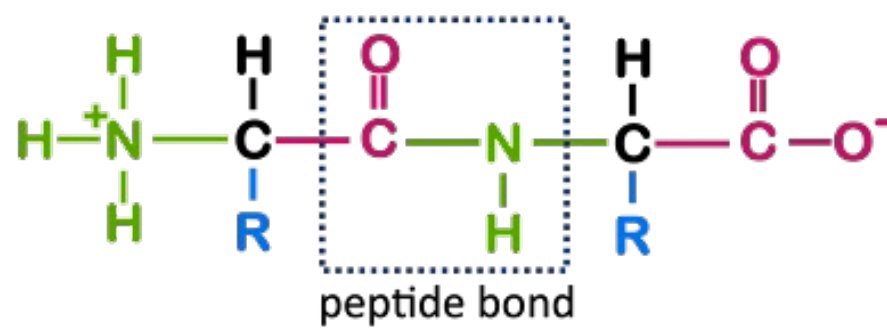
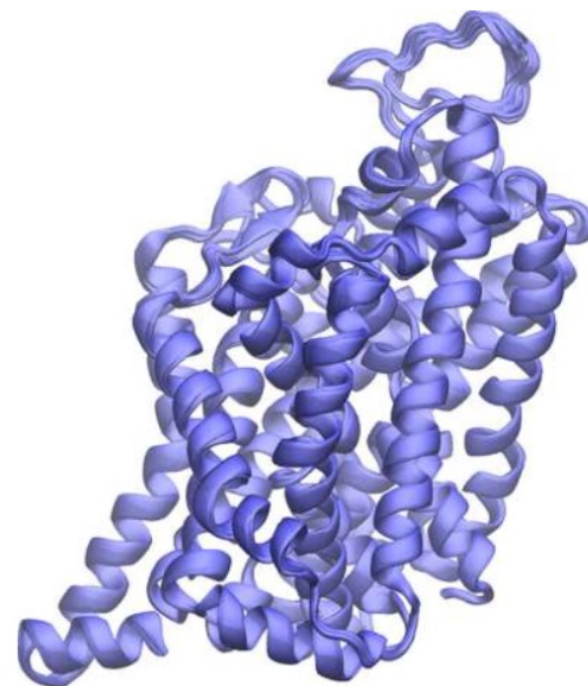
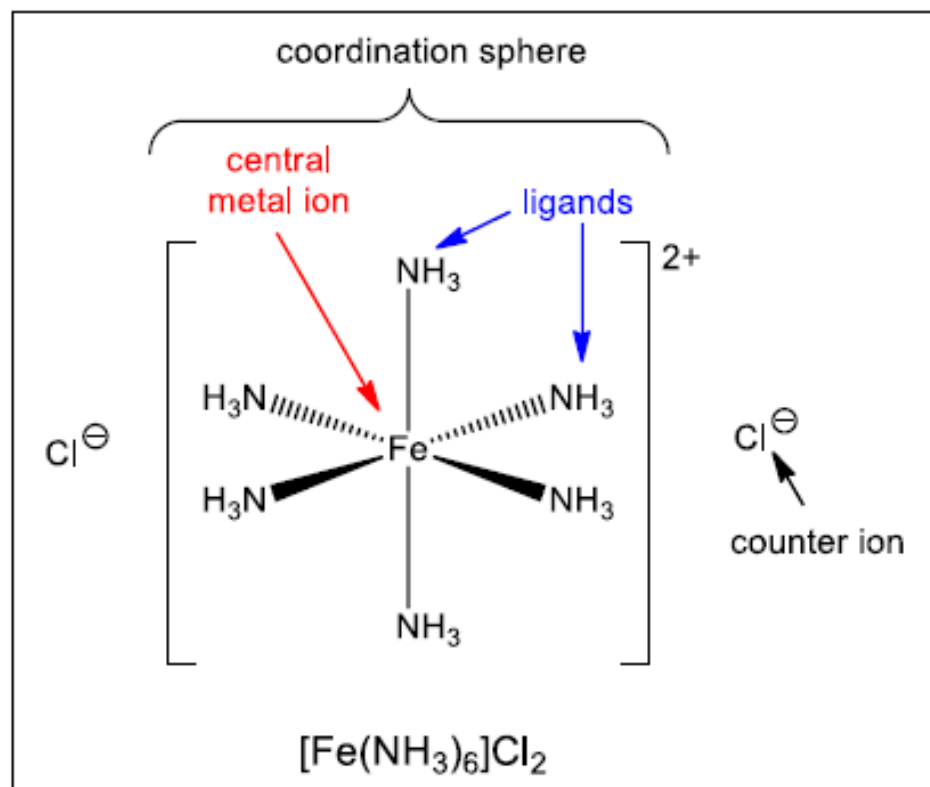
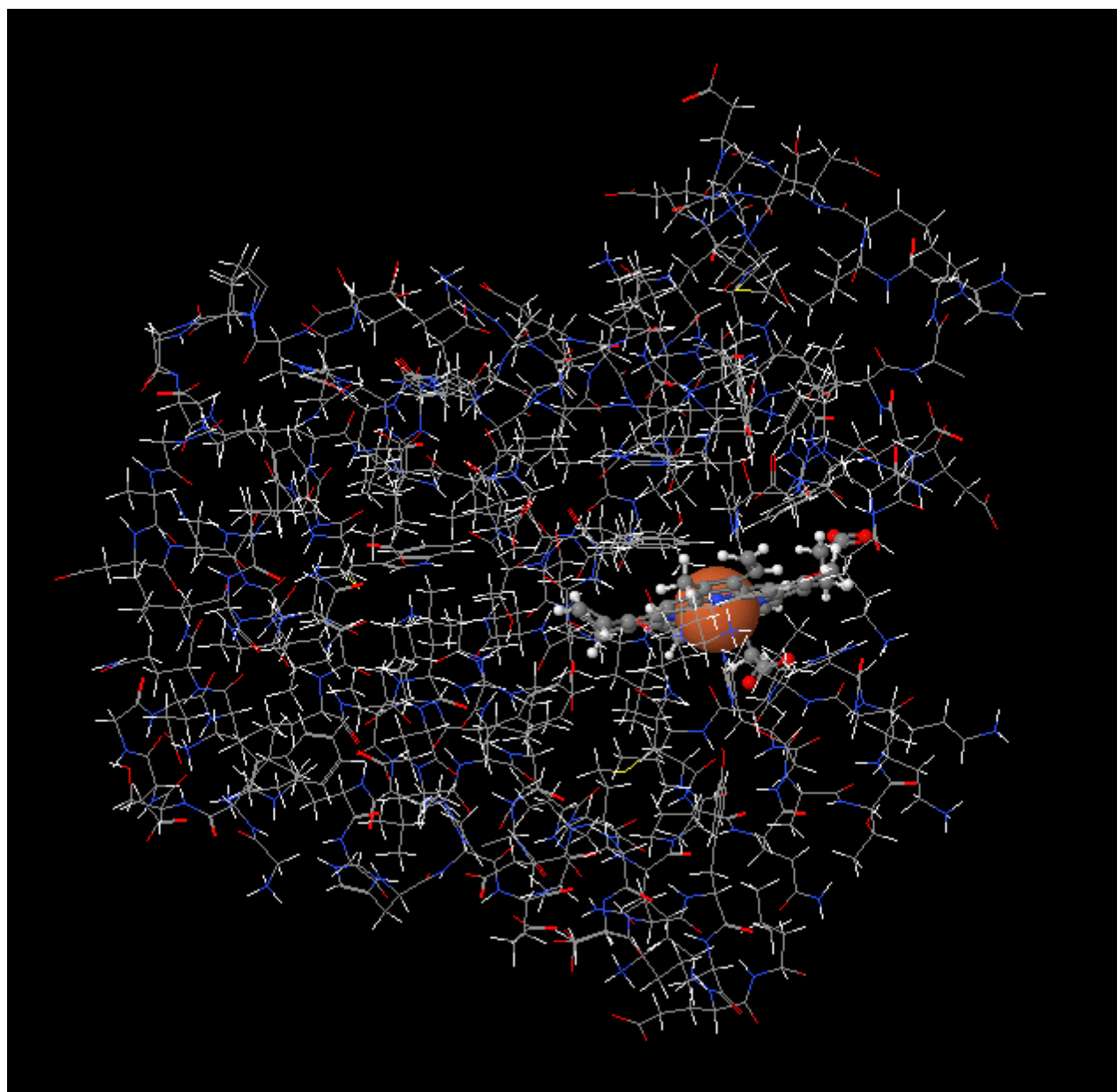
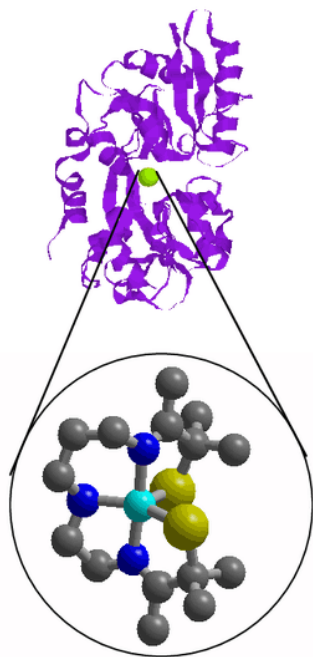


Oxygen transport and storage system
Hemoglobin
Myoglobin





Metalloproteins and Metalloenzymes

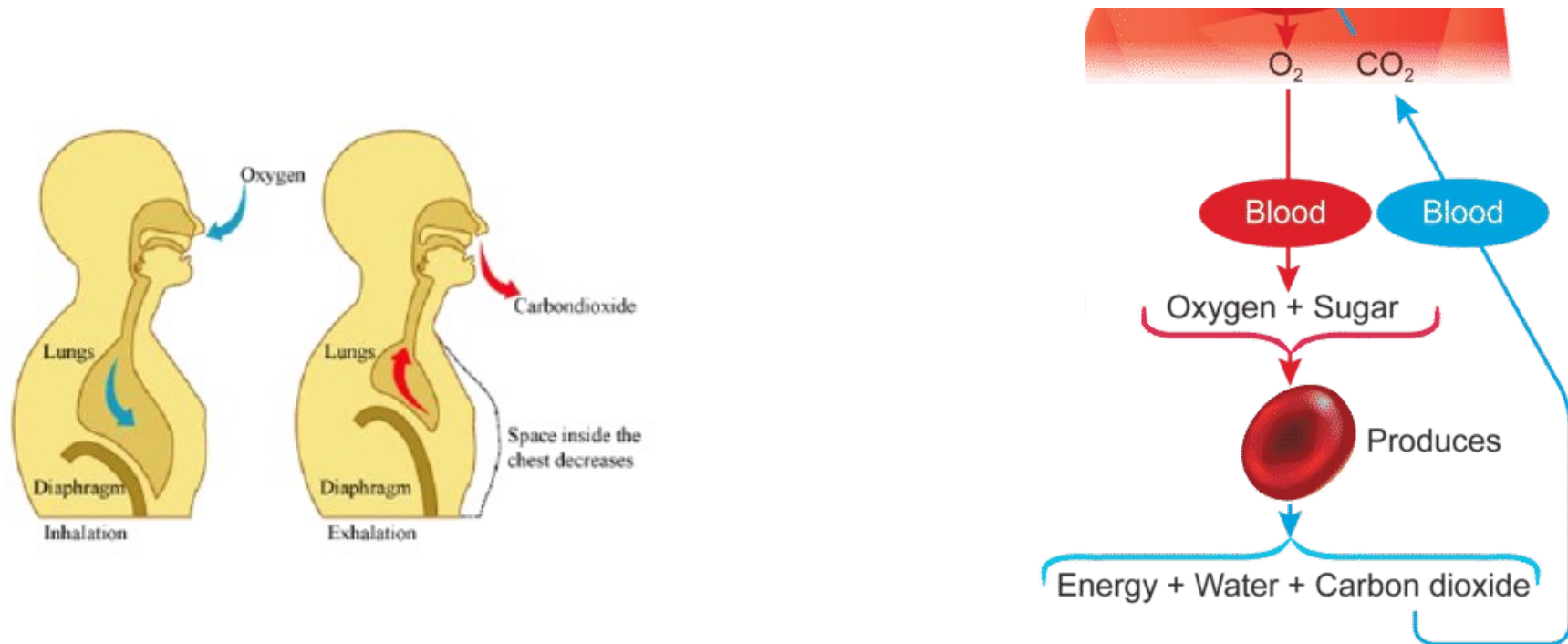


- 33% of all enzymes contain transition metal ions!
- Metal ions PROMOTE REACTIONS
 - bond cleavage
 - bond formation
 - electron transfer
 - atom transfer
 - controlled radical rxns, etc.....
- Metal ions PROMOTE PROTEIN FOLDING

Metalloproteins are proteins that contain **tightly bound metal ions** (at least one) eg. Zn, Cu, Fe. Mostly bound by N, S, O atoms of the protein residues

Metalloenzymes: A subset of metalloproteins. One of the coordination site is labile. The chemistry of metals allows a broader set of reactions and enzymatic actions

Oxygen needs to be transported from **LUNGS** to **Tissues** via **Blood** to generate Energy



How is oxygen transported?

Can blood transport oxygen?

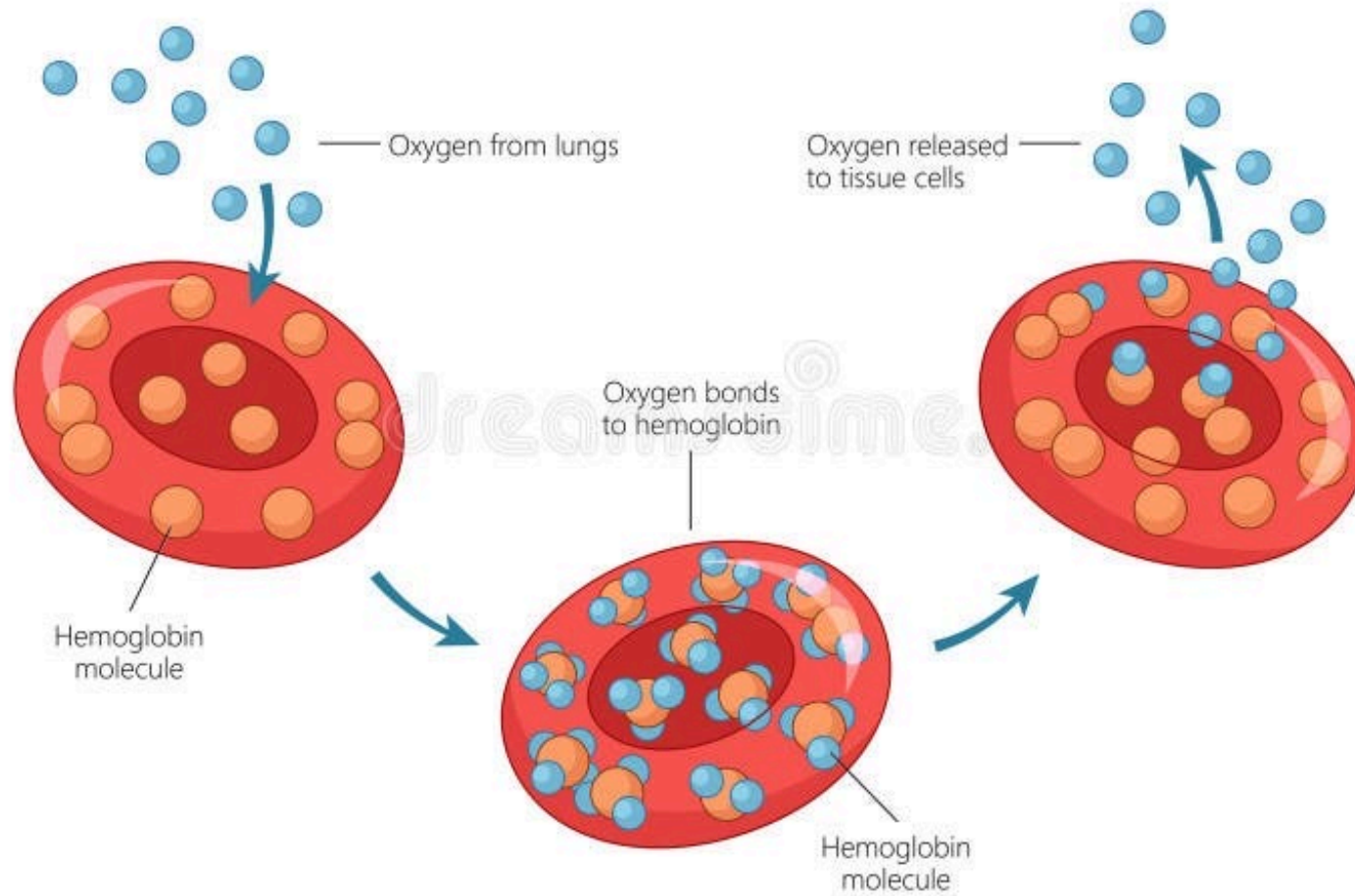
Oxygen solubility in pure or fresh water at 25 °C and 1.0 atm of O₂ pressure is about $1.22 \times 10^{-3} \text{ mol dm}^{-3}$

Oxygen solubility in human plasma is about $10^{-4} \text{ mol dm}^{-3}$

Dissolved oxygen in human blood constitutes only about 1.5% to 2% of the total oxygen transported

Nature has a solution for an efficient oxygen transport system

Transport of Oxygen in the Blood

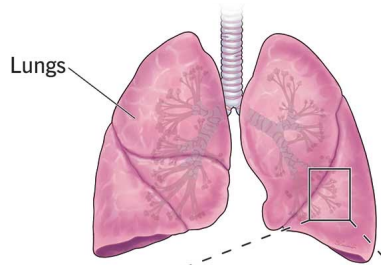


Need for a controlled system for transportation

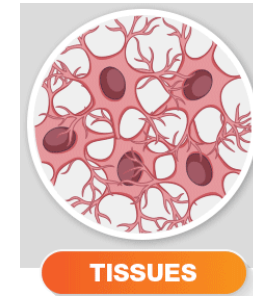
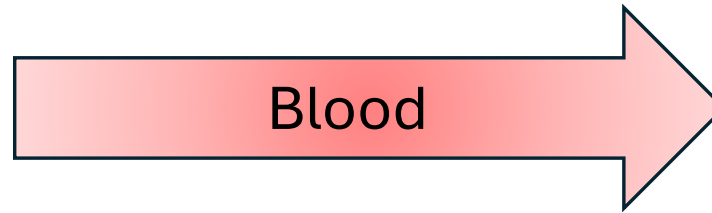
Hb needs to bind oxygen in the lungs

Hb needs to release oxygen in the tissues

Hb = Hemoglobin



Lungs



Tissues

Oxygen concentration is high

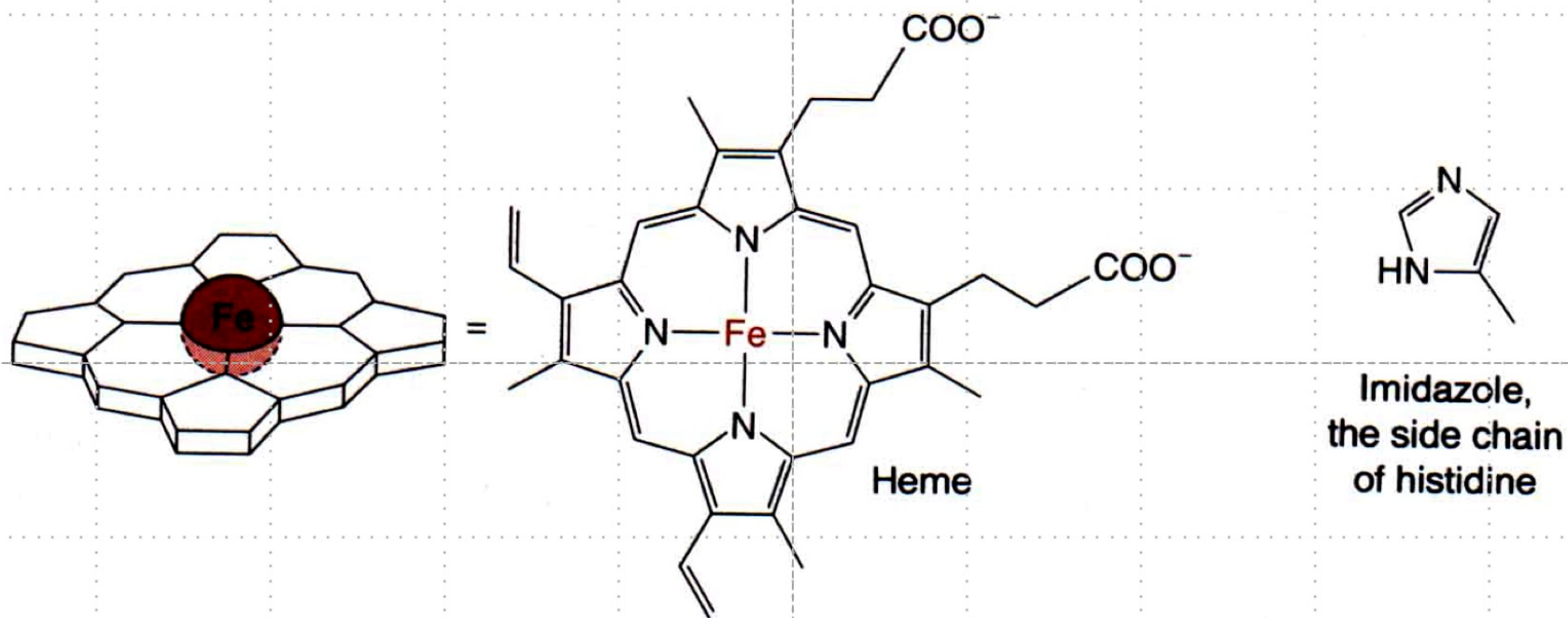
pH is high (~7.4)

Oxygen concentration is low

pH is low (~7.0-7.2)

Metabolism → Lactic acid production

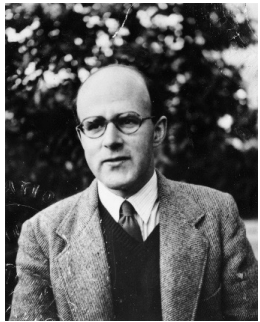
The Heme Group; the Defining Example of a Bioinorganic Chip



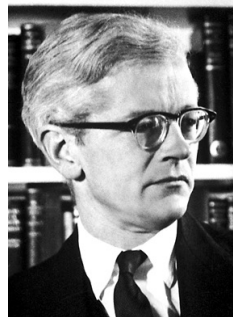
Peripheral carboxylates and axial ligands matter!

Hemoglobin: Metalloprotein for O₂ transport

Nobel Prize 1962

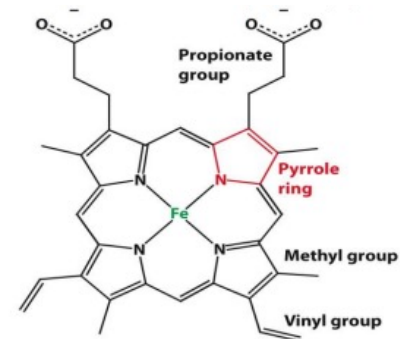
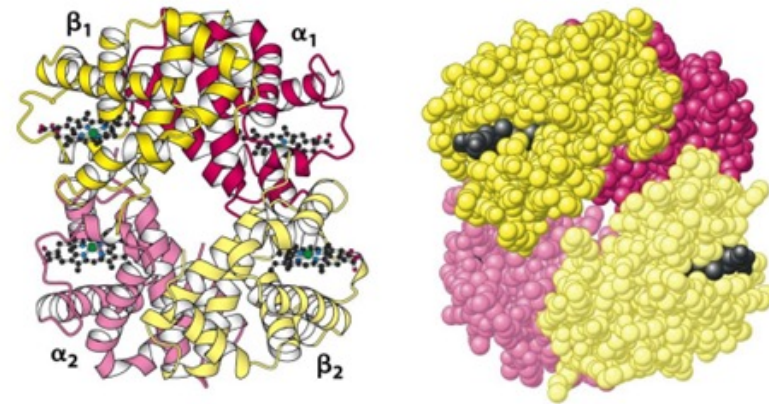


Max Perutz



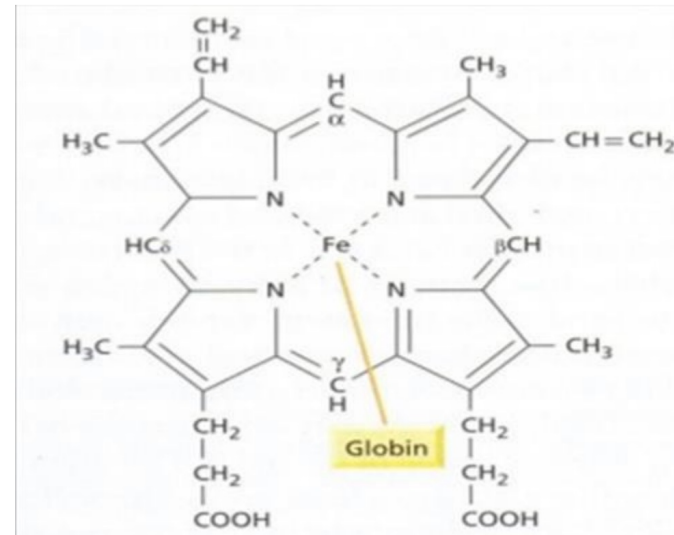
John Kendrew

Perutz, M. F. *et al. Nature* **185**, 416–422 (1960)
Kendrew, J. C. *et al. Nature* **185**, 422–427 (1960).



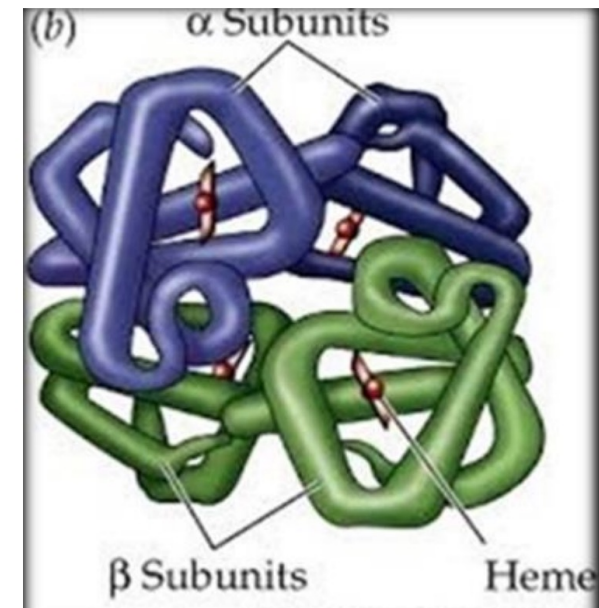
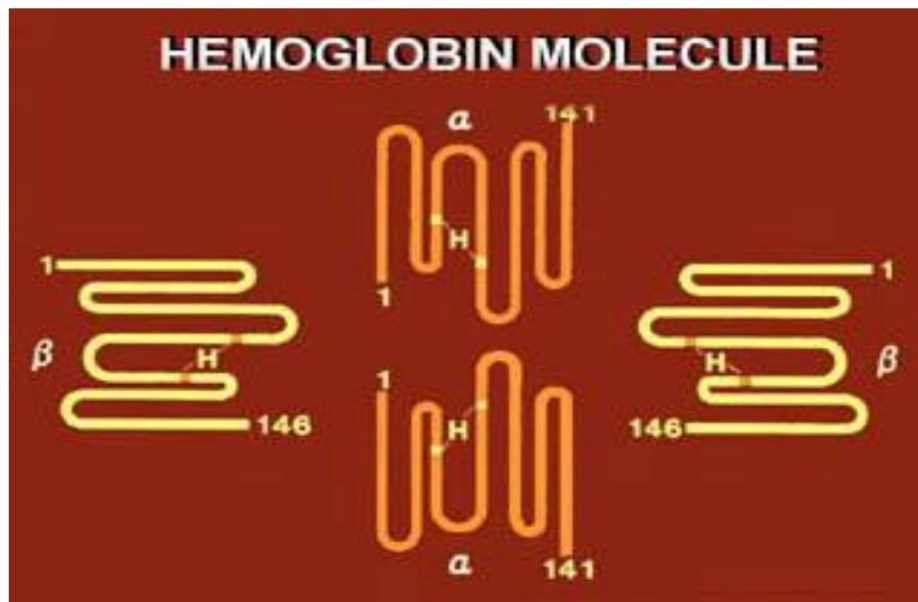
Structure of hemoglobin

- Iron containing pigment **heme** is attached to protein **globin**
- **Heme** is iron porphyrin complex called **Iron protoporphyrin IX**
- **Globin** is a protein



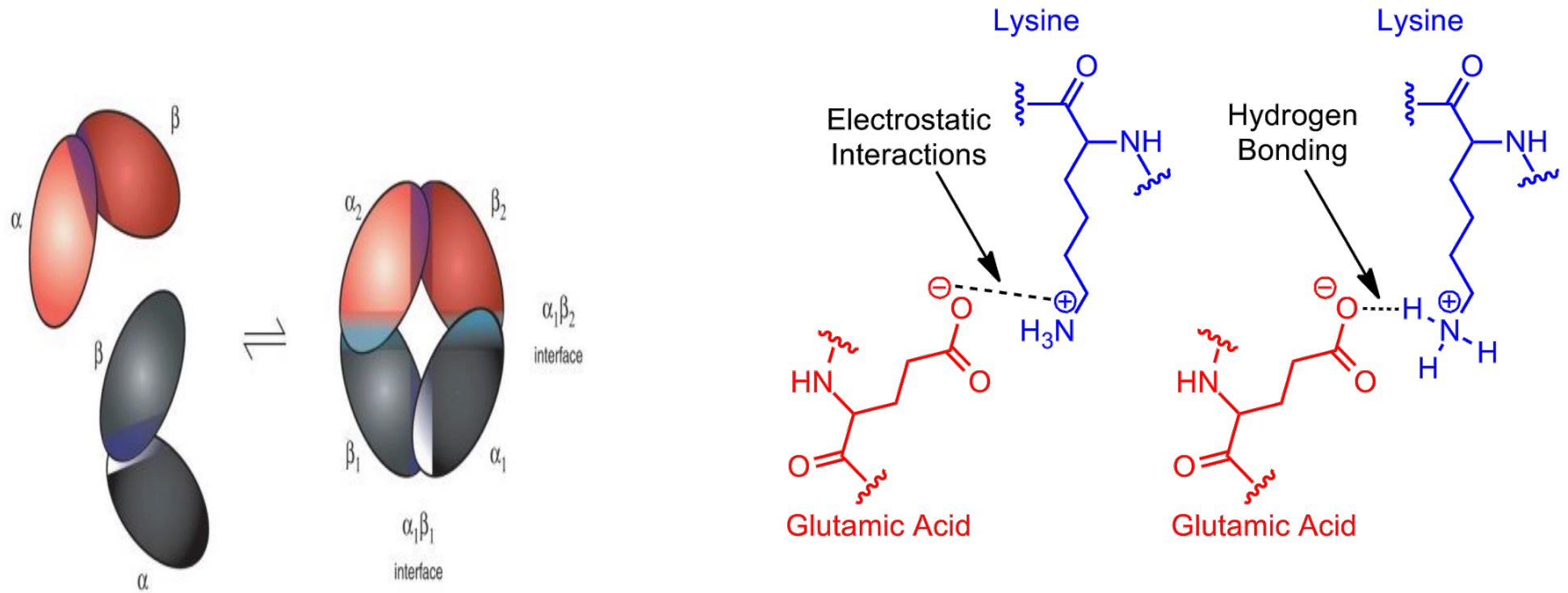
Structure of Globin

Adult Hemoglobin has 4 Polypeptide chains : **2 α** and **2 β**



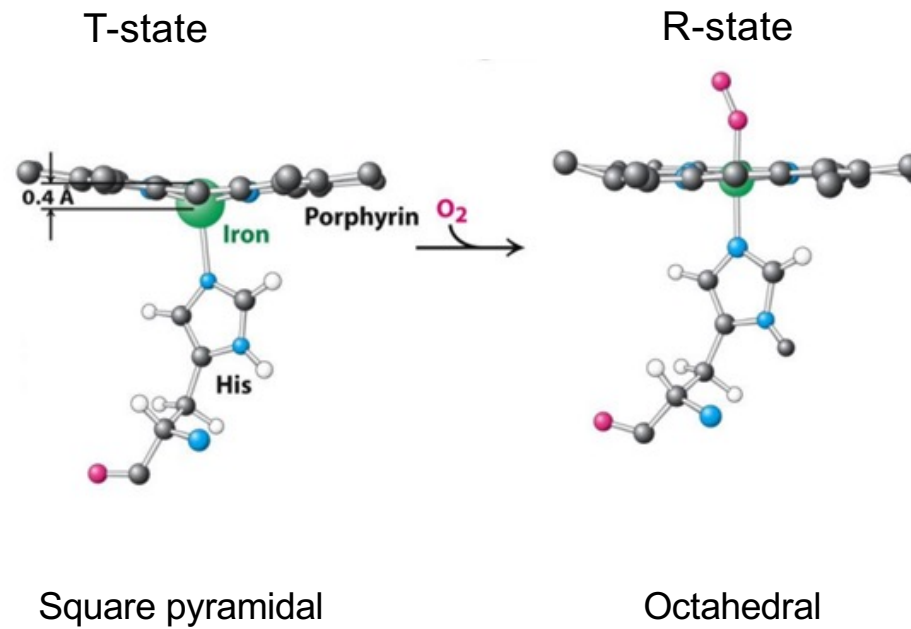
Four units: Tetrameric

Hb Globin: Four Units Linked by Salt-Bridge Interactions



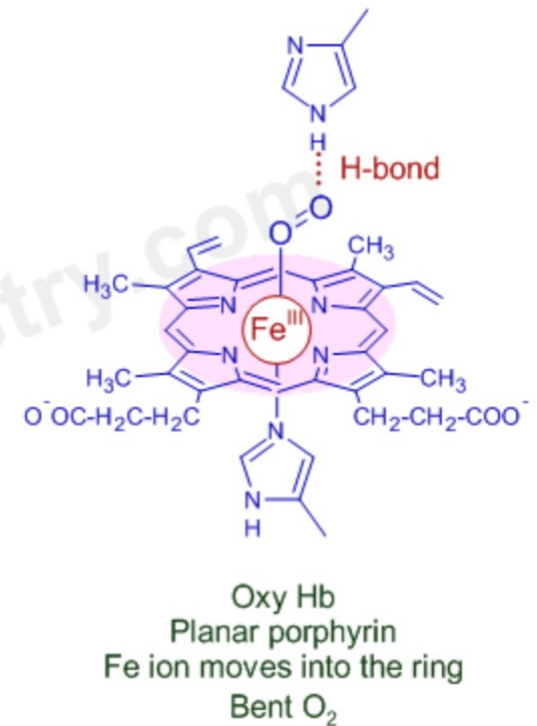
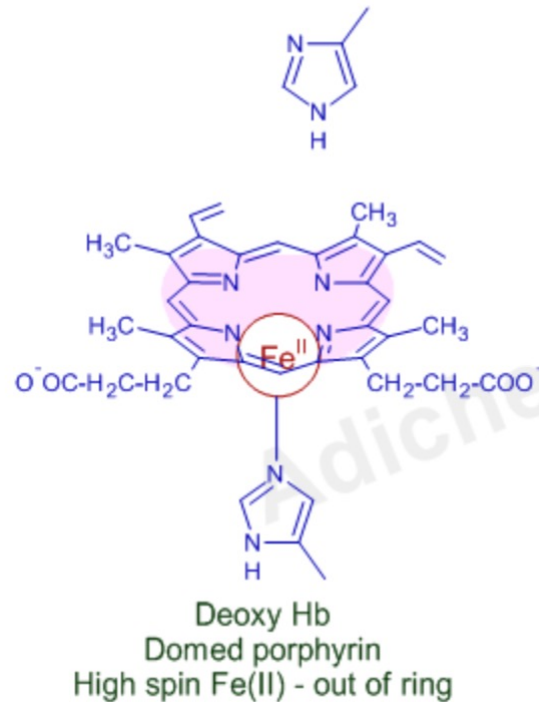
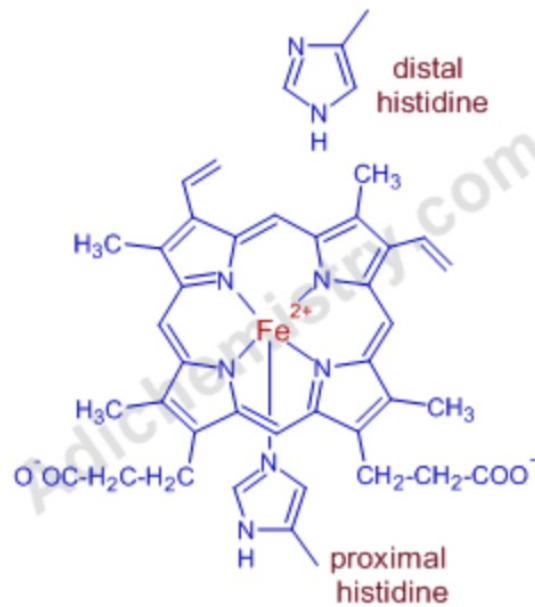
In chemistry, a salt bridge is a combination of two non-covalent interactions:
hydrogen bonding and ionic bonding

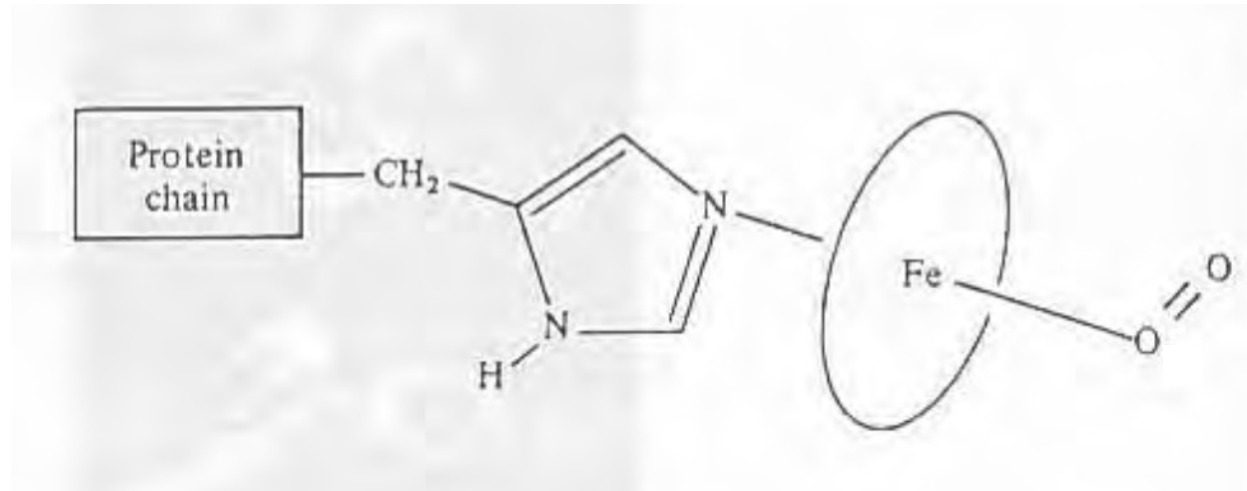
Shape of heme changes on oxygen binding



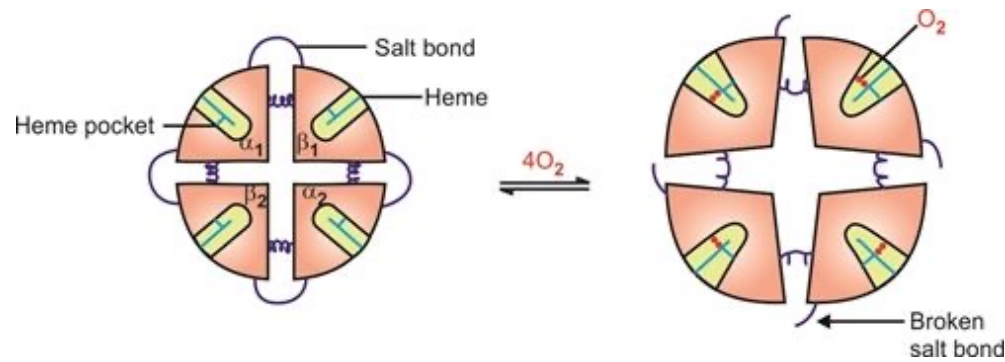
Fe²⁺ oxidation state in both T and R form

Structures of active site of Oxy and Deoxyhemoglobin



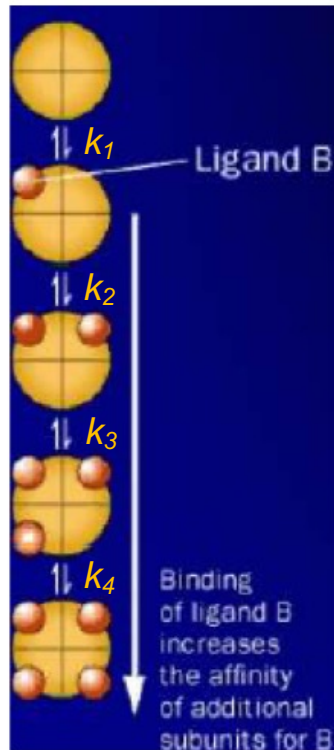


Due to change in shape, Salt Bridges are broken between the dimer units



T Form of Hb	R Form Of Hb
Deoxy Hb is in T form binds with CO ₂ , H ⁺ and 2,3BPG	Oxy Hb is in R form binds only with Oxygen
T form has 8 salt bridges linked in between the dimer subunits	Salt bridges are broken in between the dimer subunits during oxygenation of Hb.
More constrained form	Less constrained form
T form has low affinity for Oxygen	R form has higher affinity for Oxygen
T form of Hb predominates in low pO₂	R form of Hb predominates at high pO₂

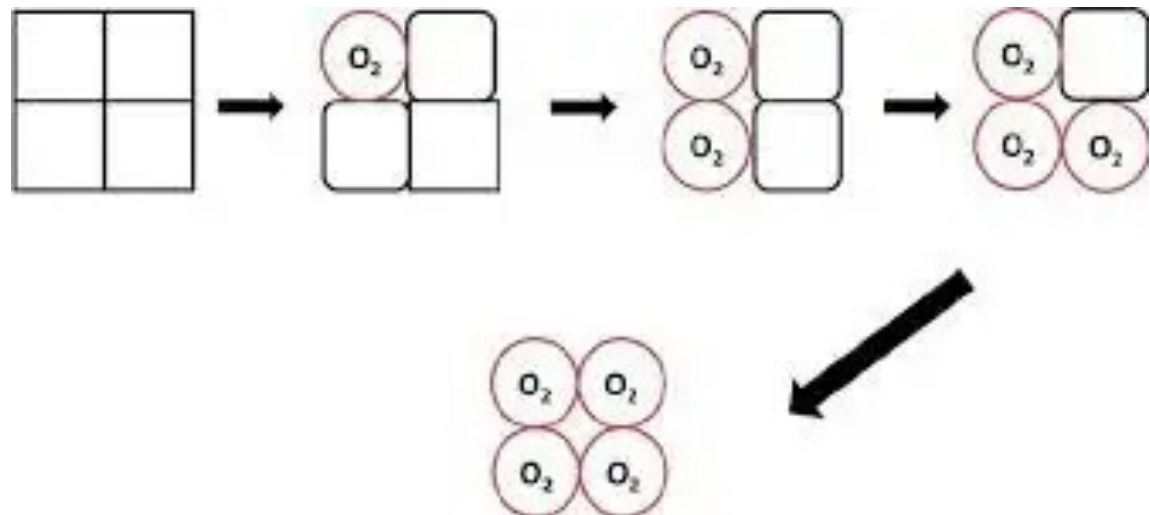
Cooperative Binding

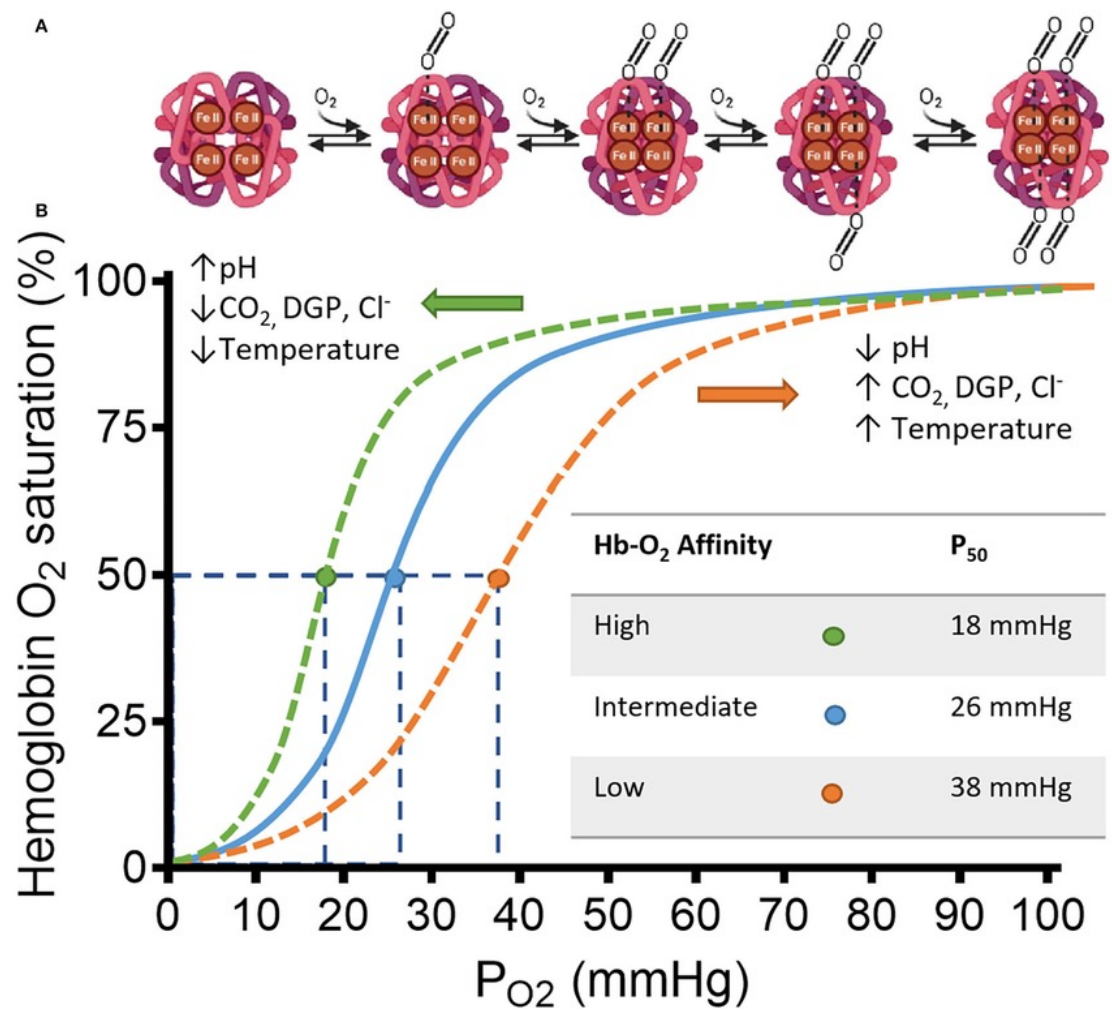


$$K_4 > K_3 > K_2 > K_1$$

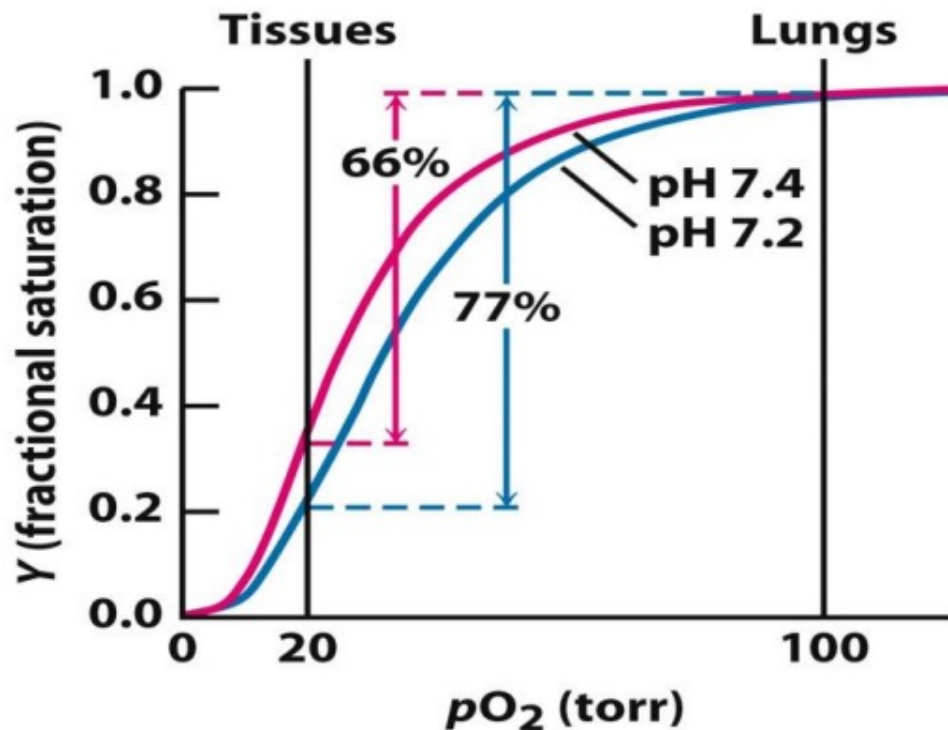
The affinity of O_2 increases as O_2 binds

Cooperative binding in hemoglobin is the process where oxygen binding to one heme group increases the affinity of the other heme groups for oxygen, allowing hemoglobin to efficiently pick up and release oxygen, characterized by an S-shaped binding curve (**sigmoidal**)



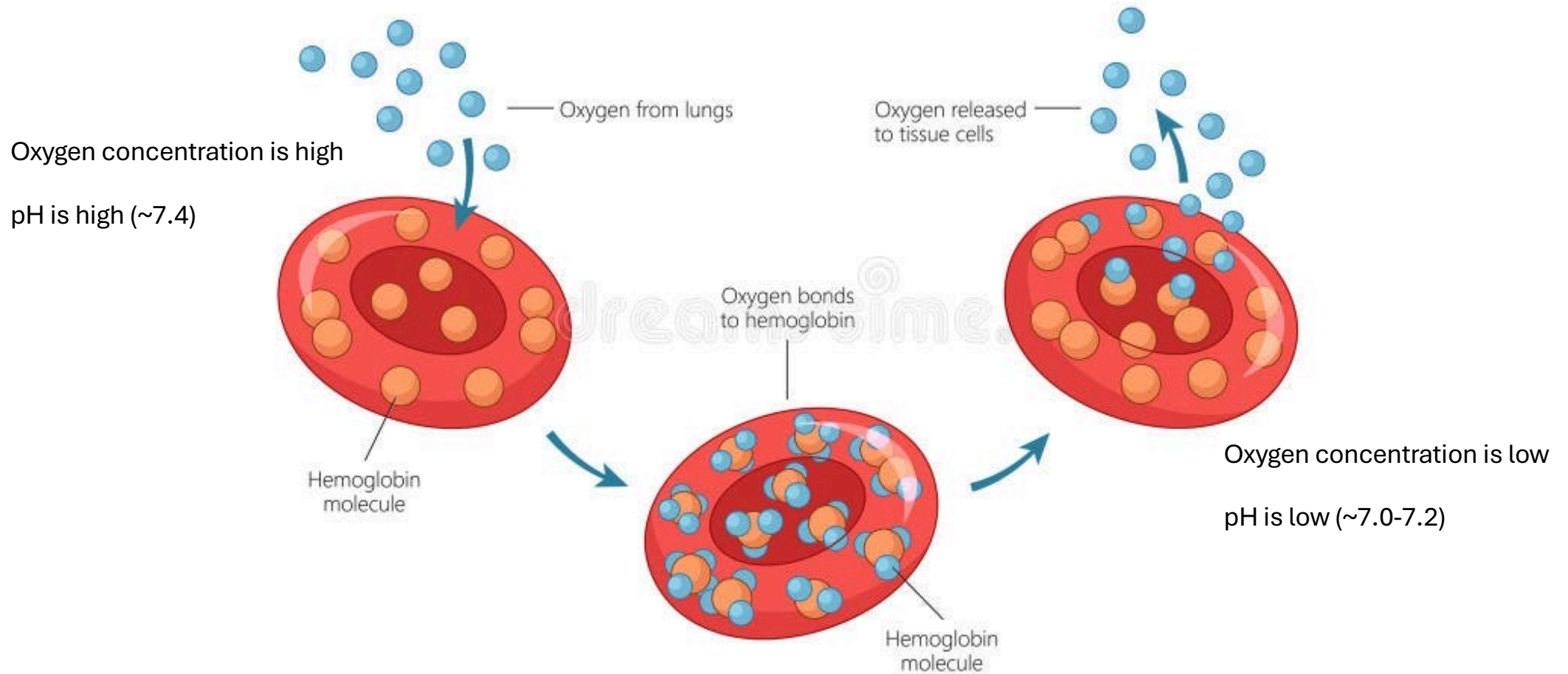


Bohr effect



The Bohr effect describes how hemoglobin's ability to bind oxygen decreases as pH drops (becomes more acidic) and carbon dioxide levels rise, prompting it to release oxygen, especially in active tissues where these conditions occur; conversely, in the lungs, lower CO_2 and higher pH promote oxygen binding, ensuring efficient gas exchange by shifting the oxygen-hemoglobin dissociation curve.

Transport of Oxygen in the Blood

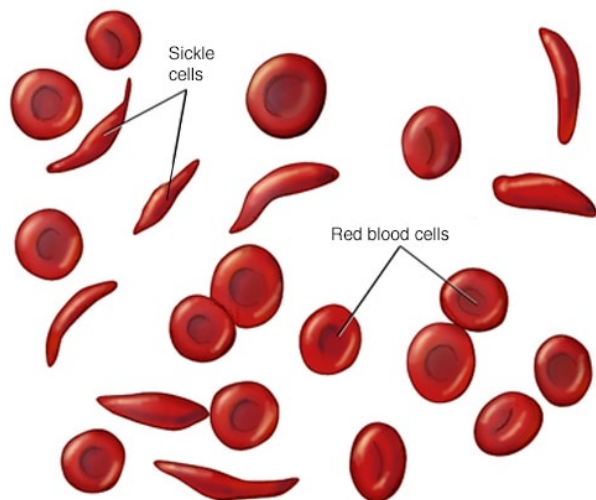


Loss of Hemoglobin function

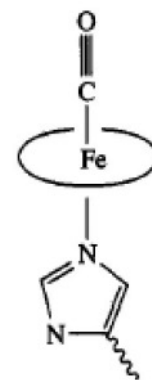
Iron deficiency: Anemia and Cardiovascular strain

Improper Hemoglobin synthesis: Thalassemia

Defective red blood cell: Sick cell anemia



Asphyxia: Insufficient oxygen



Strong π acceptors: CN, CO, PF_3

CO binds Hb ~200 times tighter than O_2

Loss of Hemoglobin function

Porphyria: Defective heme biosynthesis



Possible connection to vampire folklore

Porphyrin build up in the body

Urine red in color

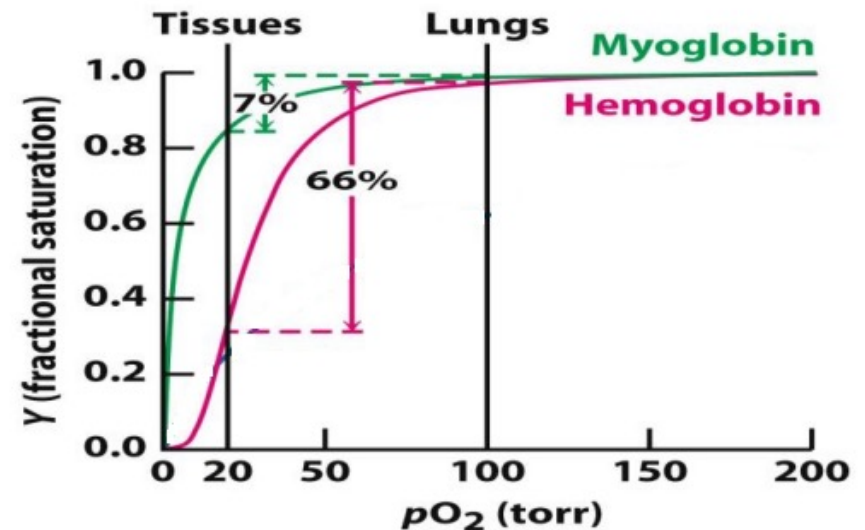
Sensitive to sun exposure

Other deadly diseases:, Thalassemia

Myoglobin (Mb): Metalloprotein that stores Oxygen



Structurally, myoglobin is equivalent to a single monomeric unit (subunit) of hemoglobin



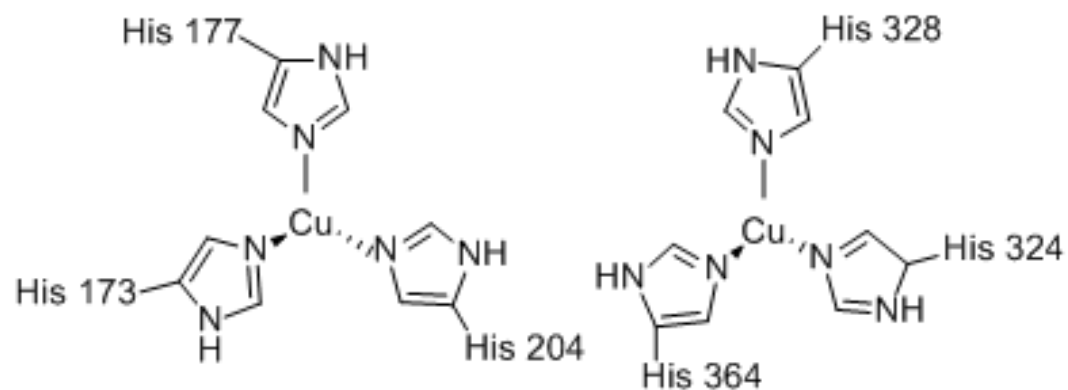
Higher affinity for oxygen than Hb
Parabolic curve rather than sigmoidal
Does not have cooperative binding like Hb



Octopus having copper based blue blood

Other examples include Lobsters, crabs, pillbugs, shrimp, crayfish, scallops, barnacles, snails, small worms (except earthworms), clams, squid, slugs, mussels, horseshoe crabs, most spiders.

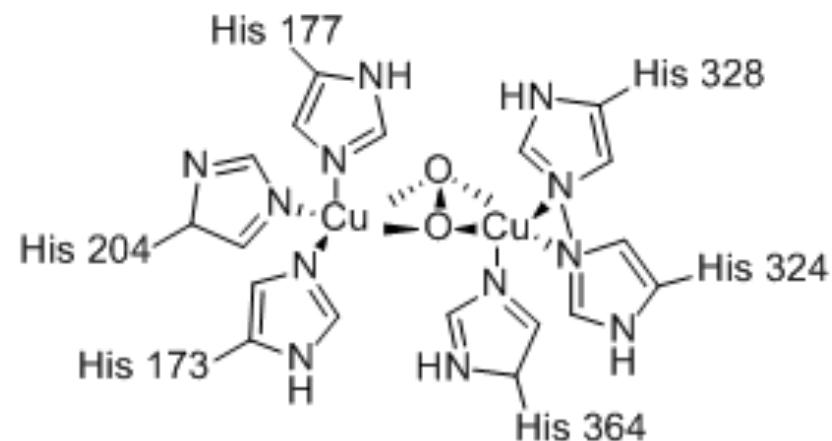
Hemocyanin



Deoxyhemocyanin

Cu(I), colorless

Cu-Cu distance – 3.7 Å



Oxyhemocyanin

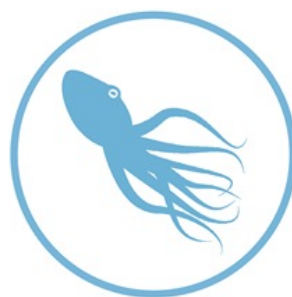
Cu(II), blue

Cu-Cu distance – 3.60 Å

Coordination chemistry behind the color blood



Red



Blue

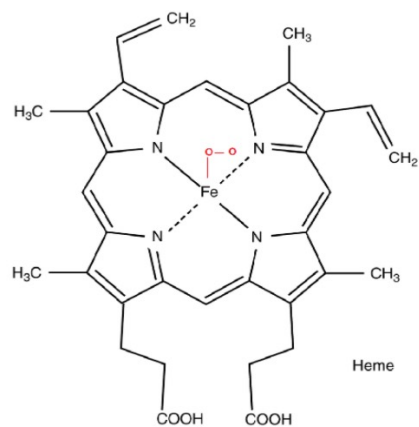


Green



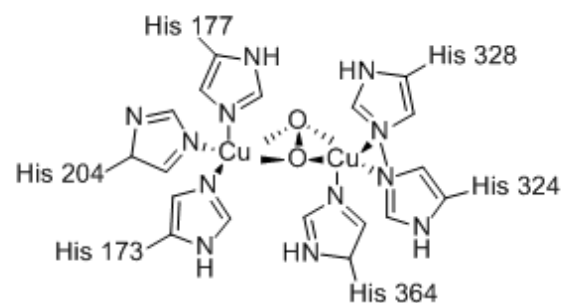
Violet

Hemoglobin



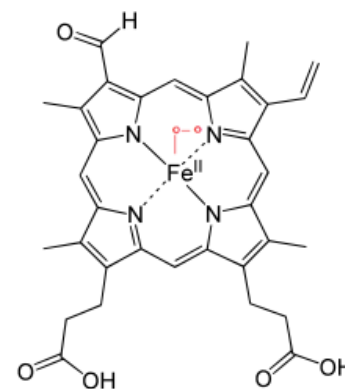
Humans and animals

Hemocyanin



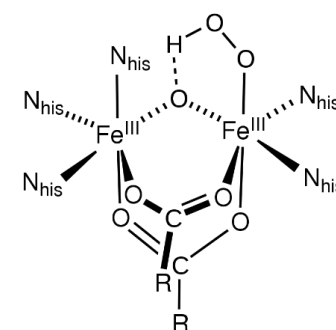
Spiders, octopus, squids

Erythrocruorin



Segmented worms, leeches

Hemerythrin



Marine worms, brachiopods